RAJALAKSHMI ENGINEERING COLLEGE

P Saam 251801276-2025-AIDS

P2

Started on	Thursday, 27 August 2026, 9:02 AM
State	Finished
Completed on	Thursday, 27 August 2026, 9:03 AM
Time taken	55 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Divide and Conquer / Binary Search approach to find the first zero
int findFirstZeroIndex(int *arr, int low, int high, int m) {
    int first_zero = m; // Default to m if no zero is found

    while (low <= high) {
        int mid = low + (high - low) / 2;

        if (arr[mid] == 0) {
            first_zero = mid;    // Record this zero position
            high = mid - 1;      // Search left for an earlier zero
        } else {
            low = mid + 1;       // Search right
        }
    }
    return first_zero;
}

int main() {
    int m;
    // Read the size of the array
    if (scanf("%d", &m) != 1) return 0;

    int *arr = (int *)malloc(m * sizeof(int));

    // Read the array elements
    for (int i = 0; i < m; i++) {
        scanf("%d", &arr[i]);
    }

    // Find the index of the first occurrence of 0
    int first_zero_index = findFirstZeroIndex(arr, 0, m - 1, m);

    // Number of zeroes is total size minus the starting index of the first zero
    int zero_count = m - first_zero_index;

    // Print the result
    printf("%d\n", zero_count);

    // Free allocated memory
    free(arr);

    return 0;
}
```

	Input	Expected	Got	
✔	5 1 1 1 0 0	2	2	✔
✔	10 1 1 1 1 1 1 1 1 1 1	0	0	✔
✔	8 0 0 0 0 0 0 0 0	8	8	✔
✔	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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